

There are three constraints (different from the non negativity constraints), so that there are three dual variables. Their sign is defined by the summary table.

Constraint of the primal	Dual variable
$2x_1 + x_2 \geq 10,$	$\gamma_1 \geq 0,$
$x_1 + 4x_3 = 16,$	$\gamma_2 \in \mathbb{R},$
$x_2 - x_3 \leq 5,$	$\gamma_3 \leq 0.$

There is a dual constraint for each primal variable. As $x_1 \geq 0$, we have

$$2\gamma_1 + \gamma_2 + 0\gamma_3 \leq 1,$$

where 2, 1 and 0 are the coefficients of x_1 in the primal constraints, and the 1 at the right hand side is the coefficient of x_1 in the objective function. The sign of the constraint is determined by the summary table. As x_1 must be non negative in the primal, the corresponding constraint in the dual is $\leq$. Similarly, as $x_2 \in \mathbb{R}$, we have

$$\gamma_1 + \gamma_3 = -4,$$

and as $x_3 \leq 0$, we have

$$4\gamma_2 - \gamma_3 \geq 2.$$

Therefore, the dual problem is:

$$\max 10\gamma_1 + 16\gamma_2 + 5\gamma_3 \quad (1)$$

subject to

$$\begin{aligned} 2\gamma_1 + \gamma_2 &\leq 1, \\ \gamma_1 + \gamma_3 &= -4, \\ 4\gamma_2 - \gamma_3 &\geq 2, \\ \gamma_1 &\geq 0, \\ \gamma_2 &\in \mathbb{R}, \\ \gamma_3 &\leq 0. \end{aligned} \quad (2)$$

In order to calculate the Lagrangian, we write the problem as

$$\min x_1 - 4x_2 + 2x_3$$

subject to

$$\begin{aligned} h_1(x) &= 16 - x_1 - 4x_3 = 0, & (\lambda) \\ g_1(x) &= 10 - 2x_1 - x_2 \leq 0, & (\mu_1) \\ g_2(x) &= -5 + x_2 - x_3 \leq 0, & (\mu_2) \\ g_3(x) &= -x_1 \leq 0, & (\mu_3) \\ g_4(x) &= x_3 \leq 0. & (\mu_4) \end{aligned}$$

The Lagrangian function of this problem is

$$\begin{aligned} L(x_1, x_2, x_3, \lambda, \mu_1, \mu_2, \mu_3, \mu_4) &= x_1 - 4x_2 + 2x_3 + \lambda(16 - x_1 - 4x_3) \\ &\quad + \mu_1(10 - 2x_1 - x_2) + \mu_2(-5 + x_2 - x_3) \\ &\quad - \mu_3x_1 + \mu_4x_3 \\ &= x_1(1 - \lambda - 2\mu_1 - \mu_3) + x_2(-4 - \mu_1 + \mu_2) \\ &\quad + x_3(2 - 4\lambda - \mu_2 + \mu_4) + 16\lambda + 10\mu_1 - 5\mu_2. \end{aligned}$$

In order for the dual function to be bounded, the coefficients of x_1, x_2, x_3 have to be zero. Therefore,

$$\mu_3 = 1 - \lambda - 2\mu_1, \quad (3)$$

$$\mu_2 = 4 + \mu_1, \quad (4)$$

$$\mu_4 = -2 + 4\lambda + \mu_2, \quad (5)$$

and the dual function is

$$q(\lambda, \mu_1, \mu_2, \mu_3, \mu_4) = 16\lambda + 10\mu_1 - 5\mu_2.$$

Moreover, we must impose $\mu_1, \mu_2, \mu_3, \mu_4 \geq 0$. As μ_3 and μ_4 do not appear in the dual function, they can be eliminated.

- Combining (3) and $\mu_3 \geq 0$, we obtain $\lambda \leq 1 - 2\mu_1$.
- Combining (5) and $\mu_4 \geq 0$, we obtain $\lambda \geq \frac{1}{2} - \frac{1}{4}\mu_2$.

Therefore, the constraints on the dual variables are

$$\lambda \leq 1 - 2\mu_1,$$

$$\lambda \geq \frac{1}{2} - \frac{1}{4}\mu_2,$$

$$\mu_2 = 4 + \mu_1,$$

$$\mu_1 \geq 0,$$

$$\mu_2 \geq 0.$$

Therefore, the dual problem is written as

$$\max 16\lambda + 10\mu_1 - 5\mu_2$$

subject to

$$\lambda + 2\mu_1 \leq 1,$$

$$\lambda + \frac{1}{4}\mu_2 \geq \frac{1}{2},$$

$$-\mu_1 + \mu_2 = 4,$$

$$\mu_1 \geq 0,$$

$$\mu_2 \geq 0.$$

If we define

$$\gamma_1 = \mu_1, \quad \gamma_2 = \lambda, \quad \gamma_3 = -\mu_2,$$

we obtain

$$\max 10\gamma_1 + 16\gamma_2 + 5\gamma_3$$

subject to

$$\gamma_2 + 2\gamma_1 \leq 1,$$

$$\gamma_2 - \frac{1}{4}\gamma_3 \geq \frac{1}{2},$$

$$-\gamma_1 - \gamma_3 = 4,$$

$$\gamma_1 \geq 0,$$

$$-\gamma_3 \geq 0,$$

which corresponds to the optimization problem (1)–(2) obtained using the summary tables.